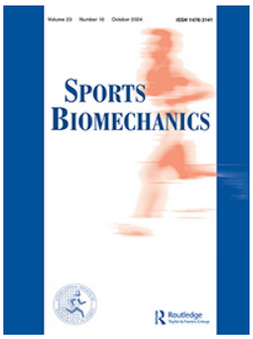

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The effect of fatiguing muscle contractions on kicking performance in experienced soccer players

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ABSTRACT

The objective of this study was to clarify the effects of fatiguing muscle contractions of the m. quadriceps femoris on kicking abilities of experienced soccer players. 16 male professional (n = 5) and amateur players (n = 11) performed kicking tests in two conditions (fatigue and control) on separate days in a randomised crossover design. The fatiguing protocol performed with the kicking leg consisted of 5 sets of 10 maximal voluntary concentric and eccentric knee extensions. Maximal voluntary isometric contraction force (MVIC), 15 hz/50 hz stimulation force ratio (force ratio), and kicking abilities were assessed before and after completion of the fatiguing protocol or rest (control). The fatiguing protocol successfully induced fatigue of 14.0 ± 2.7% (mean ± SE) reduced MVIC and 14.0 ± 3.7% reduced force ratio while no reductions occurred in the control condition. Between group difference showed ball speed declined 2.1 ± 0.95% more following the fatigue protocol compared to control condition. On the control day shooting accuracy improved by 13.3 ± 5.6% and was numerically impaired on the intervention day by 1.0 ± 9.2%. Despite this, no significant between group difference was observed in shooting accuracy (p = 0.18). The study demonstrated that fatigue induced by prior muscle contractions impairs maximal shooting speed, but we observed no significant impairment of shooting accuracy.

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KEYWORDS
Muscle fatigue; low frequency fatigue; soccer kick; kicking accuracy; kicking speed

Introduction

Soccer is the largest and most popular sport in the world (Lovett, 2018). Crucial to the outcome of a soccer match is the ability to score goals. The soccer kick has been extensively studied as a crucial element in this regard (Barfield et al., 2002). Two important components of a well-performed kick are ball speed and the accuracy with which a player is able to place the ball (Dörge et al., 2002; Katis et al., 2013; Lees & Nolan, 1998). The ability to achieve high speed and accuracy in soccer kicks may be influenced by muscle fatigue as the physical demands of soccer are characterised by intermittent high-intensity work (Bangsbo, 2014; Bangsbo et al., 2006). This intermittent work leads to both momentary fatigue during a match due to high-intensity periods as well as gradual accumulation of fatigue towards the end of the game (Dambroz et al., 2022).

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Muscle fatigue is defined by Allen, Lamb referred to as prolonged low frequency force depression or low-frequency fatigue (LFF). LFF is a phenomenon characterised by a greater decrease in muscle force production when tested at low stimulation frequencies compared to high stimulation frequencies after fatiguing muscle contractions (Keeton & Binder-Macleod, 2006). Furthermore, recovery of muscle performance is slow from LFF, and typically ranges from a few hours to several days (De Ruiter et al., 2005; Edwards et al., 1977; Jones et al., 1989). The primary cause of LFF is a reduced release of Ca^{++} from the sarcoplasmic reticulum (Allen et al., 2008). In humans LFF elicits deficits in both force and power during dynamic contraction (Herskind et al., 2023). In a soccer context, LFF has been demonstrated to be present during post-match recovery expressed as reduced 10 hz/100 hz muscle stimulation force ratio (Rampinini et al., 2011). Although fatigue of more temporary character is also present after intense periods during soccer matches (Mohr et al., 2005) we have focussed particularly on the more long-lasting LFF in the present study.

In experiments based on multi-joint movements, Forestier and Nougier (1998) examined the effect of muscle fatigue in the wrist flexor muscles on coordination and accuracy of seated throwing. Fatigue, induced by isometric contractions of the wrist flexor, reduced the proportion of accurate throws from 77% to 57%. This reduced accuracy occurred concomitantly with changes in throwing mechanics from a successive proximal-to-distal accelerating movement to a more simultaneous movement of joint systems (shoulder, elbow, and wrist) displaying increased rigidity (Forestier & Nougier, 1998). These results are interesting in a soccer context as soccer kicks in a fresh state follow similar movement characteristics as throwing (Kellis & Katis, 2007). Whether changing kinematics due to fatigue are also present during a maximal soccer kick is uncertain.

The literature on soccer kicks indicates that maximal kick speed and foot speed before ball contact are reduced in the fatigued state immediately after intense activity (Apriantono et al., 2006; R. Ferraz et al., 2012; R. M. P. Ferraz et al., 2019; Kellis et al., 2006; Torreblanca-Martínez et al., 2020). However, due to lack of objective assessments of muscle fatigue, the level of fatigue among the participants in these studies is uncertain. Some discrepancy is observed in the effect of fatigue on kicking accuracy. Thus, Torreblanca-Martínez et al. (2020) found no change in kicking accuracy after sprint work while Bjelica et al. (2013) found a decrease in soccer players' kicking accuracy kicking at different intensities (self-selected or maximum) when performed in a fatigued state. In line with this, Stone and Oliver (2009) found a decrease in points scored in the Loughborough Soccer Shooting Test (LSST) after 45 minutes of soccer-like interval-based muscle work, accompanied by an 85% increase in kicks resulting in 0 points (worst outcome of a kick).

Kicking accuracy is often measured by the LSST which divides the football goal into scoring zones (Alkhawaldeh, 2022; Engler et al., 2023; Gardasevic & Bjelica, 2019; Islam et al., 2019; Stone & Oliver, 2009; Torreblanca-Martínez et al., 2020). The test is argued to be valid and does also discriminate between groups with expected differences in skill level (Ali et al., 2007). However, the test is limited by a high complexity, low spatial resolution of scoring zones, and participants' risk willingness as scoring zones with high points and zones with low points are adjacent (Dambroz et al., 2022). To eliminate this methodological issue a simpler

'distance to target' kick-test is proposed to measure kicking accuracy (Bjelica et al., 2013; R. M. P. Ferraz et al., 2019; Finnoff et al., 2002; Radman, Wessner, Bachl, Ruzic, Hackl, Baca, et al., 2016).

Examining the effect of muscle fatigue in this context poses further challenges. These include the rapid recovery of acute muscle fatigue after running or soccer match simulation. Therefore, the results from such protocols are highly dependent on the time interval between the fatigue protocol and the test protocol (Bjelica et al., 2013; Dambroz et al., 2022; R. M. P. Ferraz et al., 2019; Russell et al., 2011; Torreblanca-Martínez et al., 2020). Another problem arises from the fact that indirect measures of fatigue are diverse and could include central cardiovascular parameters such as heart rate (R. Ferraz et al., 2012, 2017; Kellis et al., 2006; Radman, Wessner, Bachl, Ruzic, Hackl, Prpic, et al., 2016; Russell et al., 2011; Stone & Oliver, 2009), perceived exertion (RPE) on the Borg scale (R. Ferraz et al., 2012, 2017; R. M. P. Ferraz et al., 2019; Radman, Wessner, Bachl, Ruzic, Hackl, Prpic, et al., 2016; Russell et al., 2011), or surrogate measures such as blood lactate concentration (R. Ferraz et al., 2012, 2017; R. M. P. Ferraz et al., 2019; Kellis et al., 2006; Radman, Wessner, Bachl, Ruzic, Hackl, Prpic, et al., 2016; Russell et al., 2011). None of these parameters directly refer to Allen and Lamb's definition of muscle fatigue (Allen et al., 2008) and only RPE has been shown to yield a negative correlation with kicking accuracy (Torreblanca-Martínez et al., 2020; Vøllestad, 1997). To eliminate issues regarding the time sensitive nature of acute muscle fatigue and to align with the definition of muscle fatigue (Allen et al., 2008), measuring maximal voluntary isometric contraction force (MVIC) and force induced by electrical stimulation with high or low frequency after performing a fatigue protocol would be advantageous. LFF has a long duration and is relatively stable during the first hour after a fatigue protocol when soccer tests are completed (Keeton & Binder-Macleod, 2006; Sieljacks et al., 2015).

The purpose of the present study was to clarify the effect of muscle fatigue on kick performance among skilled soccer players. The hypothesis was that a fatiguing protocol inducing LFF and decreased MVIC would lead to a reduction in maximum ball speed, maximum foot speed, and kicking accuracy of the soccer players.

Materials and methods

Study design

The present study is a randomised controlled crossover trial comparing the effects of a fatigue protocol on soccer kick performance to a control condition (rest). Group 1 (n = 9) received the fatiguing protocol on the first trial day and then the control condition separated by one week. Group 2 (n = 7) received the control condition first and then the fatiguing protocol separated by one week (see Figure 1 for illustration of study design and testing). At a visit three days prior to the first trial day, participants were familiarised with tests in the dynamometer and electrical stimulation.

The study was conducted in accordance with ethical research guidelines and participation was voluntary and was consented to by all participants after written and oral information of the study procedures was provided. Full consideration for ethical approval was, however, waived by the regional committee for research ethics (reference number 1-10-72-124-22) as the study was not considered as biomedical research.

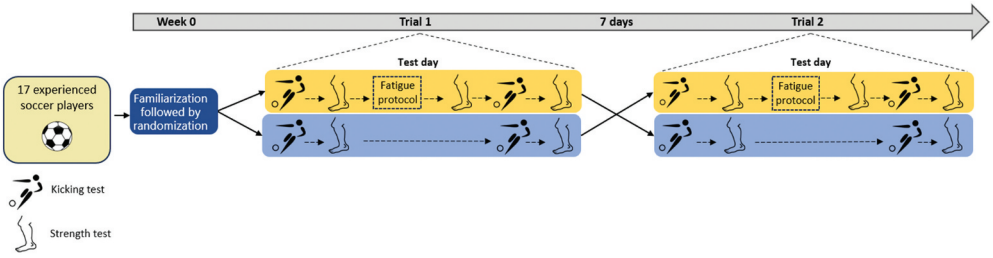

Figure 1. Overview of the experimental design and test protocol.

Participants

Seventeen experienced male soccer players ranging from high-ranking amateur (Serie1) to professional (Danish Superliga) level were recruited directly through the authors' network, advertisements in soccer clubs, and social media postings and were randomised into two groups. Group 2 experienced one dropout after the introduction day due to an ankle injury unrelated to the current study. To be eligible for inclusion participants had to be active male soccer players aged 18–30 years with a minimum of 10 years of experience of playing organised soccer (Apriantono et al., 2006), and currently play at a Serie1 level or higher. Further, participants were encouraged to refrain from physical activity 48 hours before testing. Descriptive characteristics of the participants are presented in Table 1.

Table 1. Descriptive characteristics of the participants. Data are presented as mean ± SD. Groups were pooled as no differences in characteristics were found between the two groups.

	N	Age (years)	Height (cm)	weight (kg)	Experience (years)
Participants	16	24.6 ± 2.1	184,5 ± 5.7	80.3 ± 5.8	19.8 ± 2.2

Dynamometric assessment of voluntary and electrically induced contractions of the kicking leg

Participants were positioned in the isokinetic dynamometer to perform knee extension. Two 5 × 10 cm electrodes (Valutrode Lite, Axelgaard, Denmark) were placed on shaved and disinfected skin over the quadriceps muscle 15 cm above the basis of the patella and 15 cm below the anterior superior iliac crest based on the methods described by Gaemelke et al. (2021). The axis of rotation of the dynamometer was aligned with the knee, participants were secured with shoulder and hip harnesses, and range of motion of each participant was set for knee extension and flexion.

MVIC of the knee extensors was measured at a 70° knee angle (where 0 represents full knee extension) due to peak torque being greatest at around this knee angle (Pincivero et al., 2004). The hip angle was 80° (where 0° represents full hip extension). Participants were instructed to contract their muscles as quickly and forcefully as possible, and to maintain this contraction for 2–3 seconds while standardised strong verbal motivation was provided. MVIC was measured three times on the intervention day and two times on

the control day. After the MVIC measurement, the participants had a 2-minute rest period before the electric stimulation protocol.

LFF was measured using electrical stimulation. The participants were instructed to relax as much as possible before and during the stimulation. Stimulation was delivered at 125 mA with a pulse width of 0.2 ms and a train duration of 0.8s at 15 hz and 50 hz based on results from Edwards et al. (1977). The participants had a 1-minute rest period between the stimulations (Sieljacks et al., 2015). Torque data was sampled at 1500 hz and LFF was calculated as the ratio between maximal torque at 15 hz stimulation and 50 hz stimulation.

Fatigue protocol

The fatigue protocol was performed with the kicking leg and consisted of maximal voluntary isokinetic knee extensor actions performed at 60°/s both concentrically and eccentrically as controlled by the dynamometer (Humac Norm, CSMi, Stoughton, MA). One repetition was regarded as a concentric contraction of the knee extensors followed by an eccentric contraction of the knee extensors. Range of motion (ROM) was defined from 5° below maximal knee flexion angle to 5° before maximal knee extension. Participants typically had ROM of 95° corresponding to knee flexion angle 5° to 100°. One repetition lasted approximately 3 seconds and participants were given no rest between repetitions. A total of 5 sets of 10 consecutive repetitions with a 10-second inter-set rest period were performed. Participants were instructed to exert maximal effort and were provided with standardised verbal encouragement and visual feedback of their force trace on a screen. This protocol was chosen as it has previously been shown to induce LFF (Herskind et al., 2023). After the fatigue protocol participants had a 3-minute rest period before the subsequent dynamometric testing.

Kicking performance

Kicking performance was measured on the dominant leg only. Testing was performed in an indoor biomechanical laboratory with artificial grass floor using a Select Brilliant Super ball inflated to 0.9 bar. Participants used their regular soccer boots and were asked to use the same pair for both test days due to results from Sterzing and Hennig (2008). Prior to testing, participants performed a standardised warm-up consisting of 5 minutes of self-selected intensity cycling, 20 leg swings with the kicking leg (10 in the sagittal axis and 10 in the transverse axis), and 3 warm-up kicks at self-selected intensity.

Maximum ball speed was measured using a Supido Sports Speed Radar during three kicks on a stationary ball at distance of 11 metres, emulating a penalty kick. The speed radar was placed directly behind a kicking target suspended in a net. The participants were instructed to kick with maximal force using the instep kick and rested 30 seconds between kicks. The hardest kick (ball speed) was used for data analysis of all kicking performance variables. If a player improved through all 3 kicks, additional attempts were given until the improvement ceased.

Maximum foot speed was measured using a laser system just before ball contact. Two lasers placed 3 cm apart in the kicking direction recording the time between the foot intersecting laser 1 to laser 2 yielding the speed of the foot in the final swing phase. This measurement was obtained during the maximal ball speed test.

Kicking accuracy was measured using a two-camera system determining distance from ball impact to target (a cross suspended at 1.22 metres) (R. M. P. Ferraz et al., 2019). A total of 10 kicks were performed from 11 m with 30 s rest periods between kicks (Radman, Wessner, Bachl, Ruzic, Hackl, Baca, et al., 2016). Similarly to Radman, Wessner, Bachl, Ruzic, Hackl, Baca, et al. (2016) the ball was rolled forward and kicked while moving for this test. Based on results from (Andersen & Dörge, 2011; Hunter et al., 2018), participants were instructed to kick at a minimum of 80% of their measured current maximum speed. If a kick did not reach the required speed, the kick was discarded from data analysis and the player performed a new attempt. Ball speed was measured using the speed radar as in the maximal ball speed test. The two-camera system consisted of two recording wall mounted cameras (Brio 4K Ultra HD Webcam, Logitech) spanning the field in the plane of the target cross perpendicular to the kicking direction (Figure 2). The position of ball impact was determined by triangulation using pixel values from the two cameras. The Euclidian distance between ball impact and target cross was then determined and used for analysis. The analyses were performed using MatLab R2021a, MathWorks, USA.

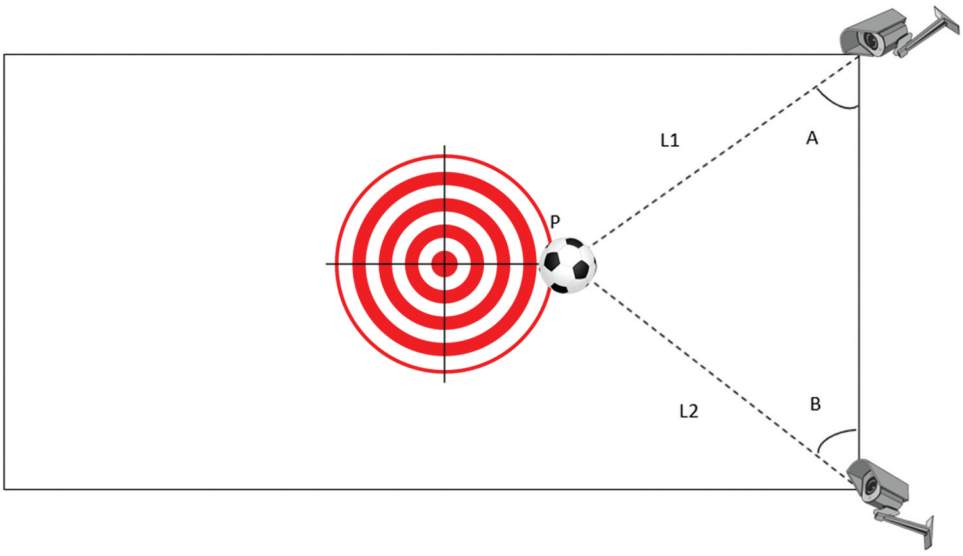

Figure 2. Illustration of the situation where camera 1 and camera 2 (C1 & C2) register a ball passing in the point P. The centre of the target indicates the target point. The angles A and B are calculated from the camera pixel outputs. The lengths L1 and L2 in the triangle are calculated to find the exact point P. The distance between target point and point P is then calculated.

Distribution of kicks

To further analyse kicking accuracy, the kick dispersion was described using bivariate distributions based on the results of Hunter (2018). A 95% CI error ellipse was obtained from each participant from the trial days. Data from left footed kickers was mirrored and pooled with data from right footed kickers. The results are described in Table 3 comparing short axis, long axis, area, and angle of the error ellipse describing the kick distribution.

Statistics and data analysis

Based on previous results for kicking accuracy, where participants kicked from 11 m, similar outcomes were expected. However, due to the participants in our study not immediately performing kicks after the fatigue protocol, the expected effect size was anticipated to be half as large as Ferraz et al (R. M. P. Ferraz et al., 2019). The standard deviation was assumed to be identical to the one from Russell, Benton (Russell et al., 2011), which was 0.16 m. With a significance level set to 5%, this resulted in an estimated requirement of 20 participants being included in the study to obtain the desired power of 80%. Based on these assumptions the final study population achieved in this study (sixteen), yielded a calculated power of 65% for the accuracy outcome. For these calculations StataBE-17, StataCorp LLC, USA was used.

The torque values obtained from the dynamometer were analysed using Matlab R2021a, MathWorks, USA. The maximum value (baseline subtracted) recorded during each contraction was included for statistical data analysis across all measurements. An example of the data from the intervention day for a participant is presented below (Figure 3).

For statistical analysis of the contraction measurements and measurements of kicking performance, StataBE-17, StataCorp LLC, USA was used. The results were compared using paired t-tests, with a significance level of 5%. Regarding the intervention condition, muscle fatigue comparisons of pre, post and end condition were analysed using one-way repeated measures ANOVA with Bonferroni correction. The between group differences were calculated by subtracting the individual pre- and post-data for each condition (control and intervention) and comparing them. Assumptions of normality were checked visually using qq-plots.

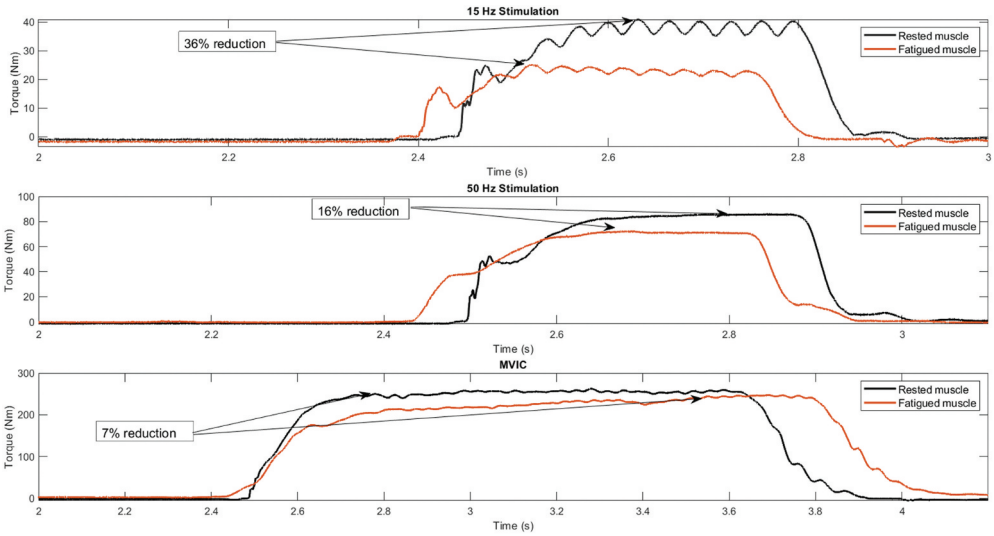

Figure 3. Example of data from the dynamometer measurements for the same participant. Here, 15Hz pre and 15Hz post (top), 50Hz pre and 50Hz post (middle), and MVIC pre and MVIC post (bottom). The black lines represent the fresh muscle (pre-measurement), and the orange lines represent the fatigued muscle (post-measurement).

Results

Seventeen experienced male soccer players (age 24.6 ± 2.1 years) ranging from amateur (Serie1) to professional (Danish Superliga) level participated in the study. The groups completed all tests during the test days, with one dropout in group 2 after the introduction day due to an ankle injury unrelated to the current study. The results from the test days were initially analysed separately for each group, and revealed no sequence effects of the experimental protocol and the effect over time was not different between the groups (not shown). Therefore, the measurements from the fatigue intervention in both groups were pooled and compared to the pooled control days. The results of the kicking performance tests from the two conditions are presented in Tables 2 and 3.

Muscle fatigue tests

MVIC measurements are illustrated in Figure 4. On the intervention day, a decrease in MVIC of 14.0% (CI = [8.7%; 19.4%], $p < 0.01$) was observed following the fatigue protocol. After the post-kicking measurements (MVIC end), the decrease was 13.5% (CI = [7.0%; 19.9%], $p < 0.01$) compared to the pre-value. There was no difference between the post-measurement and end-measurement ($p = 0.80$) showing a stable degree of fatigue. On the control day, the torque did not change at any time point, and the effect over time differed from the intervention day (12.5% CI = [2.86%; 22.2%], $p = 0.015$).

The results from the electrical stimulation are illustrated in Figures 5 and 6. We did not perform any specific measurements to ascertain that all parts of the quadriceps muscle were activated, However, we have used the present stimulation setup and electrode configuration in previous studies, where we have observed that we are able to obtain > 70% of maximal voluntary torque of knee extensors, indicating that a large part of the quadriceps is activated during electrical stimulation (Gravholt et al., 2022).

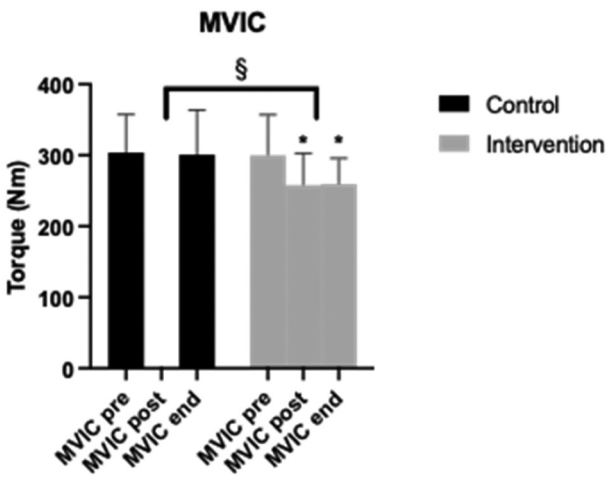

Figure 4. Torque (Nm) in MVIC (mean ± SD) from the 16 participants. *Indicates significant difference from pre-value ($p < 0.05$). § Indicates significant difference in change over time pre-end between control and intervention ($p < 0.05$). Post value in control condition was not assessed.

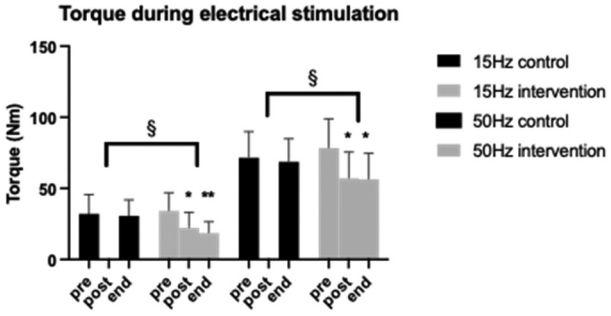

Figure 5. Torque (Nm) at 15Hz stimulation and 50Hz stimulation (mean ± SD) from the 16 participants. *Indicates significant difference from pre-value ($p < 0.05$). **Indicates significant difference from pre- and post-value ($p < 0.05$). § indicates significant difference in change over time between control and intervention ($p < 0.05$). Post values in control conditions were not assessed.

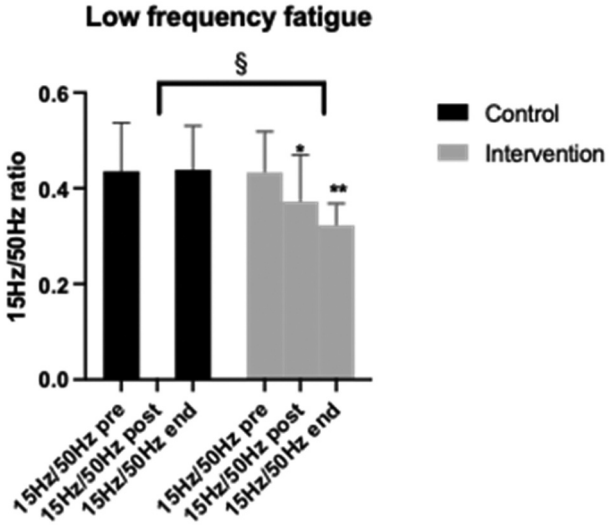

Figure 6. Ratio between torque at 15Hz/50Hz stimulation (mean ± SD) from the 16 participants. *Indicates significant difference from pre-value ($p < 0.05$). § Indicates significant difference in development over time between control and intervention ($p < 0.05$). Post value in control condition was not assessed.

During the control day no change was observed at any time point 15 hz and 50 hz stimulation. During the intervention day, the torque decreased by 35% (CI = [28.3%; 42.1%], $p < 0.01$) from pre to post for 15 hz stimulation, and further decreased 16.0% (CI = [4.7%; 27.4%], $p = 0.036$) from post to end. These reductions were significant compared to the control day (41.3% CI = [32.0%; 50.7%], $p < 0.01$). Similarly, torque at 50 hz stimulation decreased by 27.2% (CI = [18.7%; 35.6%], $p < 0.01$) from pre to post on the intervention day. No further change was found from post to end but remained decreased from pre to end 28.1%

(CI = [19.0%; 37.2%], $p < 0.01$). The decrease in torque on the intervention day was significant compared to the control day (24.4% CI = [14.3%; 34.4%], $p < 0.01$).

LFF (15 hz/50 hz ratio) is presented in Figure 6. On the intervention day, the 15 hz/50 hz ratio decreased by 14% (CI = [6.7%; 21.6%], $p < 0.01$) from pre to post and decreased further from post to end (13.3% CI = [1.9%; 24.7%], $p = 0.025$). There was no change over time on the control day ($p = 0.84$). The 15 hz/50 hz ratio reduction was significantly larger on the intervention day compared to the control day (25.3% CI = [12.3%; 38.3%], $p < 0.01$).

Kicking performance

Kicking performance is presented in Figure 7 and in Tables 2 and 3. Maximum ball speed decreased by 1.9% (CI = [0.6%; 3.1%], $p < 0.01$) on the intervention day with no effect over time on the control day ($p = 0.65$). The decrease in ball speed over time was larger on the intervention day (2.1% CI = [0.23%; 4.0%], $p < 0.01$).

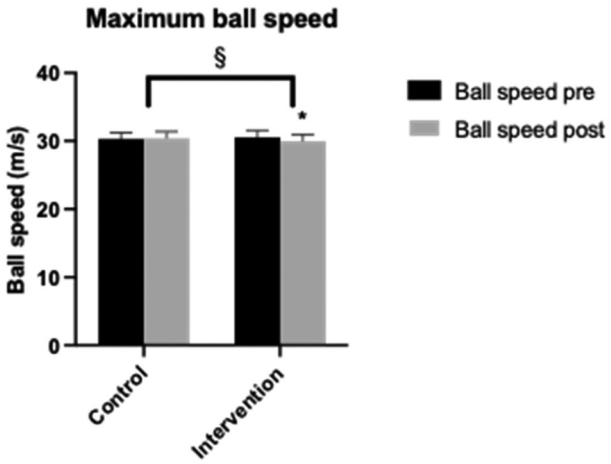

Figure 7. Maximal ball speed (m/s mean ± SD) from the 16 participants. *Indicates significant difference from pre-value (p < 0,05). § Indicates significant difference in development over time between control and intervention. (p < 0,05).

Table 2. Comparison of kicking performance results from fatigue and control condition ($n = 16$) expressed as mean ± SD or mean [95%CI] for Δ-values.

	Trial day	Before fatigue induction (pre)	After fatigue induction (post)	Δ-values [95%CI]
Ball speed [m s^{-1}]	Intervention	30.54 ± 1.01	29.97 ± 0.95	0.57 [−0.95; −0.20] *§
	Control	30.33 ± 0.90	30.40 ± 0.97	0.07 [−0.23; 0.39]
Foot speed [m s^{-1}]	Intervention	22.99 ± 3.88	21.64 ± 2.53	−1.35 [−2.50; −0.19] *
	Control	22.92 ± 3.33	23.02 ± 3.50	0.10 [−1.10; 1.28]
Foot-to-ball speed ratio	Intervention	1.33 ± 0.19	1.38 ± 0.14	0.08 [−0.11; 0.26]
	Control	1.32 ± 0.15	1.32 ± 0.16	0.01 [−0.16; 0.19]
Accuracy [m]	Intervention	1.04 ± 0.23	1.05 ± 0.21	0.01 [−0.17; 0.19]
	Control	1.05 ± 0.23	0.91 ± 0.16	−0.14 [−0.25; −0.03]*

*Indicates significant difference from pre-value (p < 0,05). § Indicates a significant difference in the change over time between groups (p < 0.05).

Maximum foot speed data is shown in Figure 8. Maximum foot speed decreased by 5.87% (CI= [0.8%; 10.9%], $p = 0.026$) on the intervention day with no effect over time on the control day ($p = 0.86$). The change in maximum foot speed over time did not differ between the days (6.30%, CI = [−0.001%; 12,7%], $p = 0.054$).

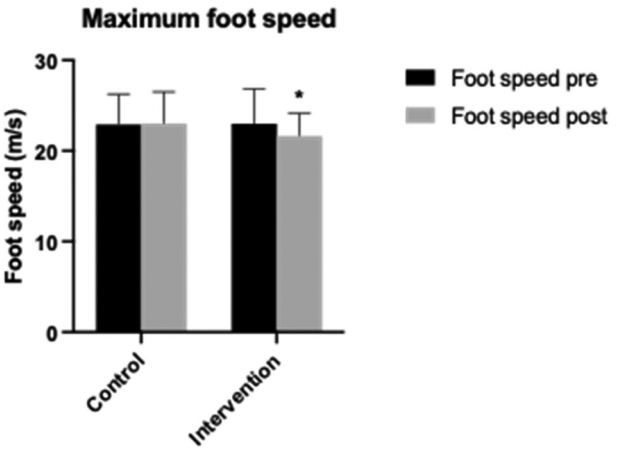

Figure 8. Maximal foot speed (m/s mean ± SD) from the 16 participants. *Indicates significant difference from pre-value ($p < 0,05$).

Shooting accuracy is shown in Figure 9. An improvement from pre to post on the control day was observed as a 13.3% (CI = [2.5%; 24.3%], p = 0.02) reduction in average distance from the target point. However, there was no change over time on the intervention day. Despite the contrasting changes on the control and intervention days, the change in kicking accuracy over time did not differ between the days (14.2%, CI = [−8.17%; 36.5%], p = 0.18).

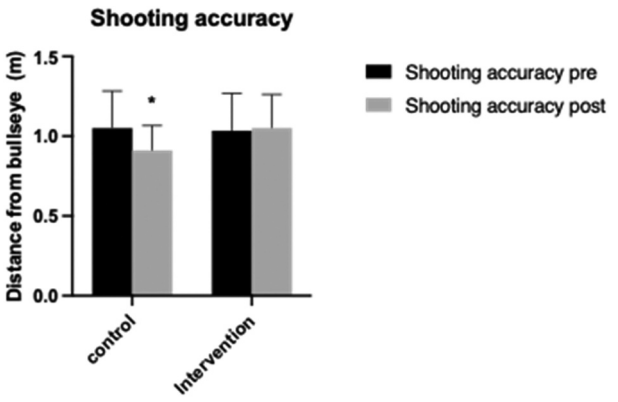

Figure 9. Kicking accuracy (distance from suspended cross, mean ± SD) from the 16 participants. * Indicates significant difference from pre-value ($p < 0.05$).

Kicking distribution is presented in Table 3 and Figure 10. No significant differences were observed in the error ellipse long axis, angle, or area. On the intervention day the error ellipse short axis increased by 53.9% (CI = [15.4%; 92.3%], $p < 0.01$). On the control day, the axis decreased numerically, and the short axis increased significantly more on the intervention day (84.6%, CI = [71.5%; 97.5%], $p < 0.01$).

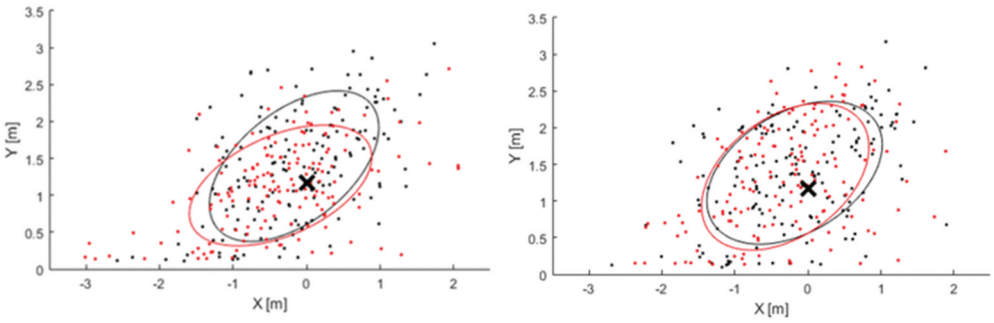

Figure 10. Kicking distribution on the control day (left) and the intervention day (right). All kicks are shown (dots) with associated average distribution pre intervention (black) and post intervention (red). The black cross indicates bullseye from the test day.

Table 3. Comparison of kicking performance results from fatigue and control condition (n = 16) expressed as mean ± SD or mean [95%CI] for Δ-values.

	Trial day	Before fatigue induction (pre)	After fatigue induction (post)	Δ-values [95%CI]
Error ellipse short axis [m]	Intervention	0.13 ± 0.08	0.19 ± 0.10	0.07 [0.02; 0.12]*§
	Control	0.15 ± 0.09	0.11 ± 0.06	−0.04 [−0.09; 0.01]
Error ellipse long axis [m]	Intervention	0.92 ± 0.40	0.90 ± 0.32	−0.02 [−0.19; 0.15]
	Control	0.97 ± 0.37	0.91 ± 0.48	−0.07 [−0.18; 0.31]
Error ellipse area [m^2]	Intervention	0.44 ± 0.58	0.49 ± 0.37	0.05 [−0.19; 0.29]
	Control	0.49 ± 0.45	0.31 ± 0.38	0.18 [−0.10; 0.46]
Error ellipse angle [rad]	Intervention	0.56 ± 0.24	0.60 ± 0.71	0.03 [−0.16; 0.22]
	Control	0.68 ± 0.24	0.51 ± 0.29	−0.17 [−0.36; 0.02]

*Indicates significant difference from pre-value ($p < 0,05$). § Indicates a significant difference in the change over time between groups ($p < 0.05$).

Discussion and implication

This study examines the effect of fatiguing muscle contractions on kicking performance among experienced soccer players. After a fatigue protocol resulting in leg extension MVIC fatigue and sustained LFF a reduction was found in maximum ball- and foot speed from an instep kick, with no changes in maximum kicking accuracy. However, in the control condition an increase in kick accuracy was found with no effect on maximum foot or ball speed. Therefore, the effect of the intervention appeared to be an absence of improvement in kicking accuracy following the fatigue protocol. Therefore, our hypothesis was partly supported. The study highlights the effect of more sustained fatigue, in this study LFF, on kicking in football since we employed a standardised fatigue protocol while using direct measures of fatigue. The

fatigue protocol elicited fatigue in m. Quadriceps somewhat similar to the fatigue experienced after a soccer match with a 14% reduction in MVIC peak force and high frequency- low frequency stimulation ratio (Rampinini et al., 2011). Most studies report a similar reduction in MVIC around 10% (3–14%) (Silva et al., 2018) examined after a soccer match with some studies finding even greater MVIC deficits at 12.7% at halftime and 18.5% at game end (Robineau et al., 2012), leaving the present reduction in MVIC in the range of post-match fatigue. Regarding low frequency fatigue Rampinini et al. (2011) reported a reduction in low frequency stimulation (10 hz) force production of 9% with no significant change in high frequency stimulation (100 hz) force production after a soccer match. Reduction in force production during low firing rates is relevant to human voluntary muscle activation as the majority of motor unit firing rates during MVIC ranges from 10-20 hz with a mean firing rate of 25 hz (Kirk & Rice, 2017). Muscle activation patterns of the quadriceps muscle during an instep soccer kick does not seem to differ much from average MVIC EMG values (Cerrah et al., 2011). Considering these similarities, we expected an effect of this type of muscle fatigue on soccer kicking performance.

In our study, the maximum kicking speed decreased by 1.9% following the fatigue protocol. Similar fatigue-induced reductions between 0% and 5% are reported in previous studies (R. Ferraz et al., 2017; Radman, Wessner, Bachl, Ruzic, Hackl, Prpic, et al., 2016; Rodríguez-Lorenzo et al., 2016; Russell et al., 2011; Torreblanca-Martínez et al., 2020). Apriantono, Nunome (Apriantono et al., 2006), used a fatigue protocol performed in a dynamometer targeting both knee extensors and flexors but with a lower intensity than the present study. They found an acute decrease of 5.6% in maximum kicking speed following fatigue. The acute nature of the test (<1 min of rest) likely contributed to a larger decrease in kicking speed compared to the present study (1.9 %) which was measured 30 minutes after the fatigue protocol. Further, Apriantono et al. (2006) reported a significant relationship between foot speed and ball speed in the fresh state which disappeared after performing muscle fatiguing work, consistent with other studies (Levanon & Dapena, 1998; Nunome et al., 2006). Such a result indicates a decrease in the variance in ball speed explained by foot speed. The reduced maximum ball speed in our study appears to occur simultaneously with a numerical decrease in maximum foot speed just before ball contact.

In our study it is assumed that the decrease in maximum foot speed during the kick explains the decrease in maximum ball speed. This is supported by the fact that the ball speed/foot speed ratio did not significantly change on the intervention day compared to the control day (−0,011 CI= [−0,086;0,064] $p = 0.75$). The results are based on the maximal trial from the participants and therefore limit the conclusions to successful trials in fatigued state.

The present study utilised direct measures of decline in muscle performance (torque) when investigating the effect of muscle fatigue on kicking accuracy. Many studies do not measure muscle performance directly before and after a fatigue protocol and therefore do not know whether or to what degree the participants are fatigued during the following tests of kicking performance (Bjelica et al., 2013; R. Ferraz et al., 2017; Kellis et al., 2006; Russell et al., 2011; W. Young et al., 2010). However, Torreblanca-Martínez et al. (2020) did use direct measures of fatigue (decrease in sprint performance) and found a 5.67% reduction on ball speed and a non-significant decrease in kicking accuracy of 7.69%.

Despite a numerical decrease in players' kicking accuracy, no significant results are found due to large variation compared to ball speed (Katis et al., 2013; Radman, Wessner, Bachl, Ruzic, Hackl, Prpic, et al., 2016). These findings are similar to the results of the present study. However, Torreblanca-Martínez et al. (2020) did not include a control group for comparison of kicking performance. Hence, limiting the possibility to detect the true effect of the fatigue protocol as the none-fatigued condition could differ from a no-change effect. This is supported by the findings of the present study and Currell et al. (2009) who found a significant improvement in shooting performance in their intervention compared with placebo. The present study showed an improvement in kicking accuracy on the control day of 13.3% ($p = 0.02$). Despite this conclusion, the significant improvement observed on the control day still comes as a surprise to us and is to our knowledge not observed before perhaps due to the lack of control conditions in many studies. Given the high level among the study participants, the standardised warm-up protocol, and the uncomplicated test protocol, we did not anticipate any change on the control day. Nevertheless, this unexpected outcome occurred hence, demonstrating the importance of including a control group in tests where an effect over time is not expected.

In studies by Ferraz et al. (2019) and Russell et al. (2011) a combination of direct and indirect measures of muscle fatigue was used. Analogous to our study, the studies do not find decreases in kicking accuracy. The studies used a decrease in running performance, heart rate, and perceived exertion on the Borg scale. The results from these studies show that heart rate and perceived fatigue increase, while the direct measures (running performance) do not change (R. M. P. Ferraz et al., 2019; Russell et al., 2011). Therefore, it is uncertain whether the participants actually experienced muscle fatigue in these studies, which could potentially explain why kicking accuracy was not affected.

The distribution of kicks from the participants in the accuracy test was described using bivariate normal distribution and showed similar characteristics to the results from Hunter et al. (2018). Accordingly, the effect of the fatiguing muscle contractions was an increased short axis on the error ellipse describing the distribution of the kicks. This is thought to be due to a disturbance in the timing of muscle activation in the kicking motion caused by the effect of fatigue on contractile properties in the involved musculature. When kicking a rolling ball this disturbance in timing may have resulted in a larger variance in the distance between support leg placement and ball in the moment of contact with the kicking leg.

Role of the quadriceps muscle

The type of fatigue that we have induced in the present study, LFF, seems to primarily decrease the muscle's ability to develop force during slow or static contractions without the same diminishing effect on maximum contraction velocity (Herskind, 2022; Herskind et al., 2022). This influence on the force-velocity curve of muscles exposed to LFF possibly affects intersegmental coordination during complex multi-joint movements like the instep kick. However, it should be noted that there is limited data on the effect of LFF on muscle contractions near maximum contraction velocity, and conclusions based on these data should be cautious (Herskind, 2022). Therefore, if the force velocity relationship of the m. quadriceps is affected during fatigue kicking conditions reducing timing of muscle activation and intersegmental coordination this

would cause a reduction in maximal foot- and ball speed as observed in the present study.

In a review by Young and Rath (2011), the authors discuss that resistance exercise induced increase in maximum kicking speed among soccer players is not achieved by targeting isolated parts of the kicking movement, e.g., the quadriceps muscle (W. B. Young & Rath, 2011). Based on this reasoning, it could be speculated that the opposite scenario is true; isolated fatigue in the quadriceps femoris muscle of the kicking leg would not affect maximum kicking speed. The results of the present study do however contradict this, as we show that performing an instep kick in a fatigued state (fatigued quadriceps muscle of kicking leg) does reduce maximum kicking speed. It should be noted though, that the review by Young and Rath (2011) primarily focuses on ankle speed immediately before ball contact. Results from the intervention day showed a 5.86% ($p = 0.035$) decrease in maximum foot speed in fatigued state and a trend observed in the between day difference of 6.3% ($p = 0.054$). The ambiguity of these results in the literature initiates a discussion on the role of knee extensors during a soccer kick. In a 2010 review, Lees et al. (2010) conclude that the angular velocities of the knee (860–1720 degrees per second) (Kellis & Katis, 2007) result in such high contraction velocities for the quadriceps femoris muscle that ankle-accelerating work from this muscle must be extremely limited. The authors also conclude that hip flexors are the prime movers in the movement, and the majority of knee extensor torque comes from thigh angular velocity during the movement (Dörge et al., 2002; Kellis & Katis, 2007; Kellis et al., 2006; Lees et al., 2010). Therefore, a fatigue protocol that affects the iliopsoas muscle could be an interesting extension of the study. These findings also highlight the importance of intersegmental coordination during the movement, as emphasised by the differences observed in the kinetics and kinematics between soccer players' preferred and non-preferred legs (Dörge et al., 2002; Nunome et al., 2006). These kinematic differences, such as increased resulting ball speed and foot speed for the preferred leg, are not due to differences in muscle strength between the legs but are kinetically attributed to increased work on the lower leg resulting from interaction torque through greater thigh angular velocity and a greater relative knee extensor muscle torque (Dörge et al., 2002; Nunome et al., 2006). Similarly, Apriantono et al. (2006) reported that muscle fatigue reduced this interaction torque acting on the proximal end of the lower leg as well as knee joint torque. These are probably not the only contributing factors affecting the resulting foot- and ankle speed. Another possible factor affected by the muscle could be the run-up speed before ball impact which was not measured in this study. Is the run-up speed slower due to muscle fatigue, this would result in less deceleration of the centre of mass before ball contact and therefore less impulse to be transferred to the foot (Potthast et al., 2010). This could possibly result in lower maximal foot and ankle speed.The fact that ball speed and foot speed before ball contact are reduced in relation to fatigue of the quadriceps femoris muscle in this study suggests a disruption in intersegmental coordination during the soccer kick, which may be caused by altered contraction dynamics disturbing the timing of muscle activation due to muscle fatigue (Herskind et al., 2022).

Methodological considerations

The sample size of the present study was based on a sample size predicting a need for 20 included participants. However, only 16 individuals completed this study hence the findings of the present study might be prone to type two errors.

Kicking accuracy was measured in kicks with a requirement of ≥ 80% individual maximum kicking speed as kicking accuracy decreases as the speed requirement increases (Andersen & Dörge, 2011; Hunter et al., 2018).

The methodological approach resulted in the exclusion of kicks falling below this threshold. We did not systematically analyse or count the excluded kicks in all participants. However, if the number of kicks below the speed-criterium is influenced, this outcome may represent valuable additional information for future studies.

Perspectives and practical applications

The results from this study indicate that maximal kicking speed is reduced with muscle fatigue and kicking distribution is altered with an increased error ellipse shorts axis. A possible effect of this is a reduced chance on goal scoring on a given kick late in the game when players experience muscle fatigue.

A goalkeeper's reach is determined by the speed of the kick when taken from the penalty spot. A given goalkeeper's maximal take-off velocity is constant, and the variable in the situation is the time it takes for the ball to travel after being kicked from the kicker's foot determining the time for the goalkeeper to travel after taking off from the ground. The faster the ball travels, the less time the goalkeeper can travel before the ball passes by and enters the goal. On the intervention day during the accuracy measurements, the players kicked on average 27.88 m/s ±1.16 m/s before fatigue and 27.44 m/s ±1.37 m/s after fatigue, providing the goalkeeper with an additional 0.0064s (1.6%) to react to the kick. The increased reaction time may extend the goalkeeper's reach during the kick and hence reduces the chance of goal scoring. The decrease in average ball speed during the accuracy test although not statistically significant followed the decrease in maximal ball speed of 1.9% corresponding to equal effort levels during the tests. On the control day average ball speed during the accuracy test did not change statistically but increased numerically from 27,69 m/s ±1.21 m/s to 27.92 m/s ±1.09 m/s an increase of 0.8% following a similar small non-significant increase of 0.2% in the maximal ball speed test again corresponding to equal effort levels during these tests.

Furthermore, the increased error ellipse describing the dispersion pattern of kicks could bring about changes in the proportion of kicks ending up in the 'scoring zone' (inside the goal but outside the reach of the goalkeeper). The areas of the dispersion ellipses were not significantly altered after fatigue in the present study. Therefore, the effect of muscle fatigue would be the altered shape of the error ellipse. Depending on the aiming point this more elongated error ellipse possibly results in a reduced number of shots ending up in the 'scoring zone' or alter the optimal point of aim more centrally in the goal.

Conclusion

The findings of the present study showed a reduction in the players' maximum kicking speed following a fatigue inducing protocol of the quadriceps femoris muscle. A decrease

in maximum foot speed immediately before ball contact is likely the cause of the lowered maximum kicking speed. The effect on kicking accuracy is not conclusive due to greater data variation and a lack of statistical power in the study. However, an improved kicking accuracy was observed on the control day, but not on the day of the fatigue intervention. The implications of muscle-fatiguing work on kicking skills are, however, not definitively described in this study and require further research.

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